

ME-1604-Heat and Mass Transfer

Dr. James Varghese School of Engineering, CUSAT

ME 1604 : HEAT AND MASS TRANSFER

Module I

Introduction to heat transfer – basic modes of heat transfer – conduction heat transfer –Fourier law of heat conduction– temperature dependence of thermal conductivity- general heat conduction equation in cartesian, cylindrical and spherical coordinates – boundary conditions – one-dimensional steady state conduction- critical insulation thickness -one-dimensional steady state conduction with heat generation -extended surface – two dimensional steady state heat conduction – conduction shape factor – unsteady state heat conduction in one dimension – lumped heat capacity system – semi-infinite solids with sudden change in surface temperature – Introduction to numerical methods in conduction.

Module II

Convective heat transfer – Newton's law of cooling – Prandtl number – laminar forced convection heat transfer from flat plates – fully developed laminar flow in pipes – turbulent forced convection – Reynolds' analogy – natural convection – natural convection heat transfer from vertical plates and horizontal tubes – condensation and boiling – film and drop wise condensation – film boiling and pool boiling – introduction to multiphase flow and heat transfer. Diffusion and convective mass transfer-Ficks law of diffusion.

Module III

Radiative transfer – electromagnetic radiation spectrum – thermal radiation –radiation properties- black body, gray body – monochromatic and total emissive power – Planck's law – Stefan-Boltzman law – Wien's displacement law – Kirchhoffs identity – shape factor- reciprocity relation – heat exchange between non black bodies; surface and shape resistances- electrical network analogy- heat transfer between parallel surfaces – radiation shields.

Module IV

Heat Exchangers: Type of heat exchangers- overall heat transfer coefficient -fouling factors -Logarithmic mean temperature difference (LMTD)- derivation of LMTD for parallel flow and counter flow heat exchangers-LMTD correction factor- effectiveness, NTU method of heat exchanger analysis- effectiveness derivation for parallel flow and counter flow heat exchangers. Design of parallel flow-counterflow-shell and tube multipass heat exchangers-condensers.

References:

- 1) Cengel, Heat Transfer Tata McGraw Hill
- 2) Holman J.P., "Heat Transfer", McGraw Hill International Students Edition
- 3) Incropera F.P. & De Witt D.P., "Fundamentals of Heat and Mass Transfer", John Wiley
- 4) Kreith F., "Heat Transfer", International Text Book Company
- 5) Gebhart B., "Heat Transfer", McGraw Hill
- 6) R.K. Rajput, "Heat and Mass Transfer", S Chand.
- 7) Venkanna, Fundamentals of HMT, PHI

Data Book (1) Heat & Mass Transfer – C.P Kothandaraman, New Age International.

Data Book (2) Heat & Mass Transfer – Domkundwar, Dhanpat Rai.

Approved data book to be specified in the question paper.

Type of Questions for University Exam

Q 1.Eight short answer questions of 5 marks with two questions from each of the four modules. (8x5 = 40 marks)

Q 2. to Q.5 : Two questions A & B of 15 marks from each modules with option to answer either A or B. (4x15 = 60 marks)

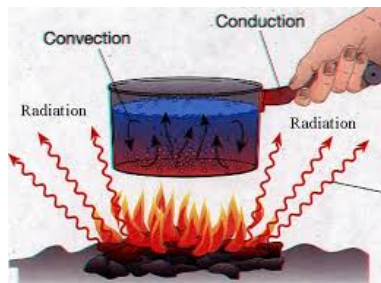
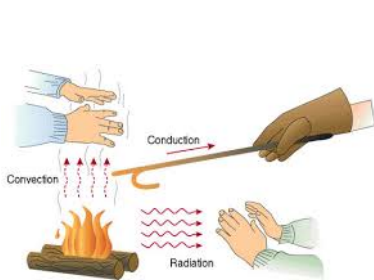
Course Outcome

- ① CO-1: To understand basic modes of heat transfer. Units, heat transfer basic applications
- ② CO-2: Analyse conduction heat transfer, one dimensional, unsteady conduction. Application of extended surfaces
- ③ CO3; Fundamentals of convective heat transfer. Forced convection problems
- ④ CO4: Analysing natural convection problems, understanding heat transfer with phase change.
- ⑤ CO5; Understanding radiation heat transfer, radiation theory
- ⑥ CO6: Radiation networks, simple radiation problems
- ⑦ CO7: Application of heat transfer, heat exchangers, design basics
- ⑧ CO8: Heat exchanger design using different methods

Introduction-Heat Transfer

- ① Heat $\implies$ Energy Transport rate (kW or kJ/s)
- ② Thermodynamics $\implies$ Relation between heat and other forms of energy
- ③ Heat transfer rate form of Energy transfer
- ④ Modes of heat transfer
 - ① Conduction
 - ② Convection
 - ③ Radiation
- ⑤ Potential $\implies$ Temperature difference (K)

Conduction heat transfer



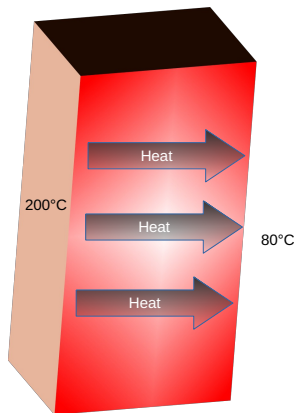
Conduction heat transfer

- 1 Heat transfer takes place as conduction in a solid if there is a temperature gradient.
- 2 conduction is mediated by the combination of vibrations and collisions of molecules
- 3 diffusion and collisions of free electrons
- 4 Conduction is within a solid or between solid objects in thermal contact.

Conduction heat transfer

Fourier law of Conduction

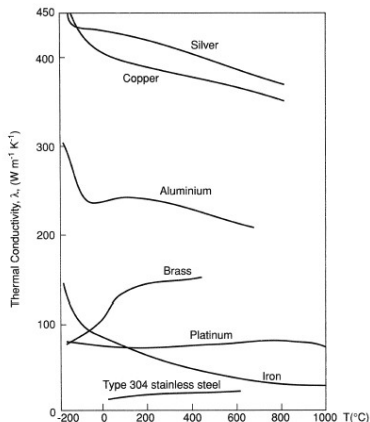
- 1 Conduction heat transfer Q (W)
- 2 $Q = -kA\frac{dT}{dx}$
- 3 k - Thermal conductivity (W/mK)
- 4 A cross sectional area through which heat transfer occurs
- 5 -ve sign heat flow direction



Thermal conductivity

Thermal Conductivity of Common Materials (at 25°C)

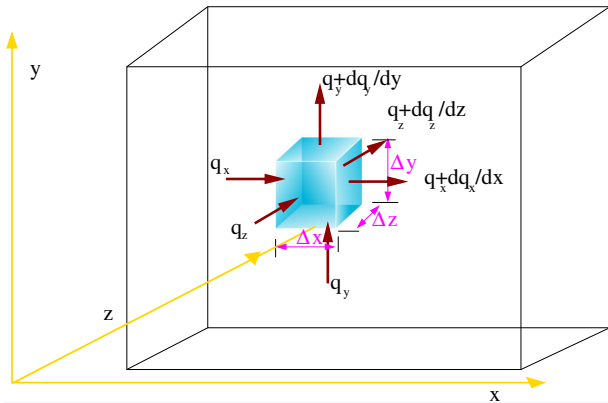
Material	Conductivity (Watts/meter-°C)
Acrylic	0.200
Air	0.024
Aluminum	250.000
Copper	401.000
Carbon Steel	54.000
Concrete	1.050
Glass	1.050
Gold	310.000
Nickel	91.000
Paper	0.050
PTFE (Teflon®)	0.250
PVC	0.190
Silver	429.000
Steel	46.000
Water	0.580



Conduction heat transfer

- 1 Less thermal conductivity materials -insulators
- 2 Conductivity changes with temperature $k(T) = k_0(1 + \beta T)$

General Heat Conduction equation-Rectangular coordinates



- 1 Elemental volume dx, dy, dz
- 2 Thermal conductivity k , density ρ , specific heat capacity c

General Heat Conduction equation-Rectangular coordinates

- 1 $T = T(x, y, z, \tau)$
- 2 Energy entering in x direction $Q_x = -kA \frac{\partial T}{\partial x}$
- 3 Energy leaving in x direction $Q_{x+\Delta x} = Q_x + \frac{dQ_x}{dx} \Delta x$
- 4 Energy leaving in x direction $= -kA \frac{\partial T}{\partial x} + -kA \frac{\partial^2 T}{\partial x^2} \Delta x$
- 5 Heat generation per unit volume $= \dot{q} \text{ (W/m}^3\text{)}$
- 6 Total heat generation $Q_{gen} = \dot{q} \Delta x \Delta y \Delta z$
- 7 Mass of the elemental volume $= \rho \Delta x \Delta y \Delta z$
- 8 Internal energy change of the element $I.E, (mc \Delta T) = \rho c \Delta x \Delta y \Delta z \frac{\partial T}{\partial \tau}$
- 9 energy balance
$$Q_x + Q_y + Q_z + Q_{gen} = I.E + Q_{x+\Delta x} + Q_{y+\Delta y} + Q_{z+\Delta z}$$
- 10 General heat conduction equation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$

General Heat Conduction equation-Rectangular coordinates

- Simplifications

- 1 General heat conduction equation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$
- 2 With no heat generation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$
- 3 Steady state conduction $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$
- 4 Two dimensional steady state conduction $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$
- 5 One dimensional steady state conduction $\frac{\partial^2 T}{\partial x^2} = 0$
- 6 Assumes conductivity and density constant.
- 7 α is the thermal diffusivity

General Heat Conduction equation-Cylindrical coordinates

1 Cylindrical coordinates

$$T = T(r, \theta, z)$$

2

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$

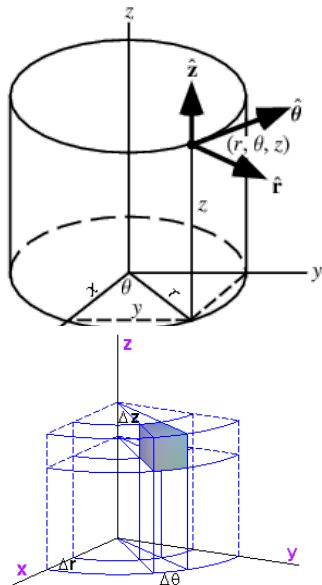
3 One dimensional steady state

$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = 0$$

4 Heat transfer in radial direction

only $Q = -2\pi krL \frac{dT}{dr}$

5 Useful to handle radial systems



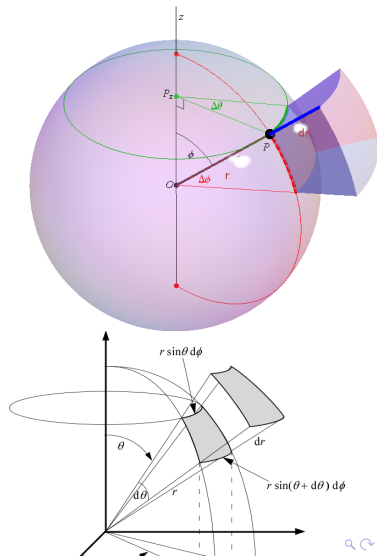
General Heat Conduction equation-Spherical coordinates

1 Spherical coordinates

$$T = T(r, \theta, \phi)$$

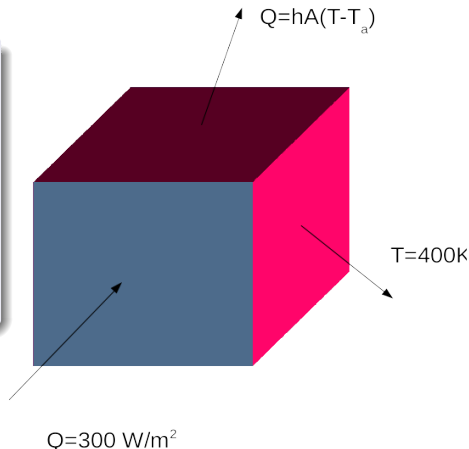
2

$$\frac{1}{r} \frac{\partial^2(rT)}{\partial r^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$



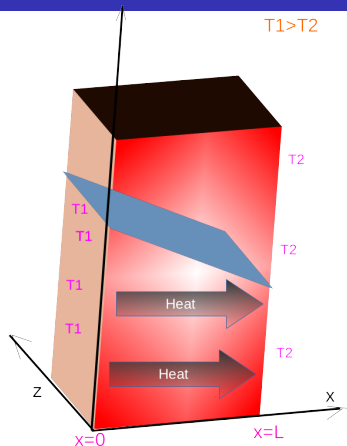
Boundary conditions

- 1 Constant temperature $T = T_1...$
- 2 Constant heat flux $q = Q/A$ (W/m^2)
- 3 Convective heat transfer
 $Q = hA_s(T - T_\infty)$
- 4 Radiative heat transfer
- 5 Combined



One dimensional steady heat transfer

- 1 One dimensional temperature variation
 $T = T(x)$
- 2 $\frac{d^2 T}{dx^2} = 0$
- 3 Integrating $\frac{dT}{dx} = C_1$
- 4 Integrating $T = C_1 x + C_2$
- 5 C_1, C_2 depends on the boundary conditions.
- 6 For b.c at $x = 0, T = T_1$ and $x = L, T = T_2$
- 7 $T = \frac{(T_2 - T_1)x}{L} + T_1$
- 8 Heat transfer $Q = \frac{kA(T_1 - T_2)}{L}$



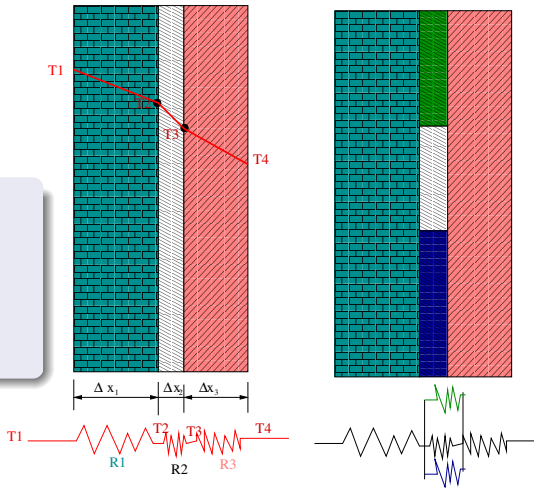
Composite walls

1 Plates in series - parallel

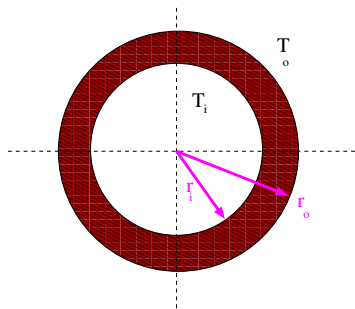
2 Same heat flow

3
$$Q = \frac{T_1 - T_4}{\Delta x_1 / k_1 A + \Delta x_2 / k_2 A + \Delta x_3 / k_3 A}$$

4
$$Q = \frac{\Delta T_{overall}}{\Sigma R_{overall}}$$

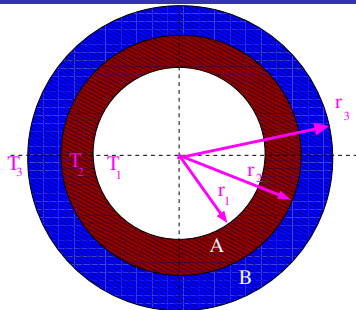


One dimensional steady heat transfer -Cylinders



$$R = \frac{\ln(r_o/r_i)}{2kl\Pi}$$

T_i ———  ——— T_o

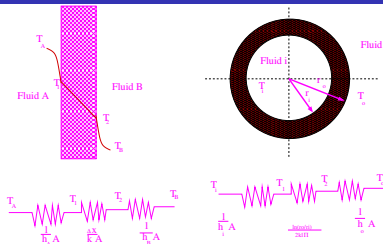


$$R = \frac{\ln(r_2/r_i)}{2kl\Pi} \quad R = \frac{\ln(r_3/r_2)}{2kl\Pi}$$

T_i ———  ——— T_2 ———  ——— T_3

- ① One dimensional temperature variation $T = T(r)$
- ② Radial heat transfer $Q_r = \frac{2\pi kL(T_i - T_o)}{\ln(r_o/r_i)}$
- ③ Composite cylinder $Q_r = \frac{2\pi L(T_1 - T_3)}{\ln(r_2/r_1)/k_A + \ln(r_3/r_2)/k_B}$

Overall heat transfer -coefficient



① Heat transfer $Q = \frac{T_A - T_B}{1/h_A A + \Delta x/k A + 1/h_B A}$, $Q = U A \Delta T_{overall}$

② Overall heat transfer coefficient $U = \frac{1}{1/h_A + \Delta x/k + 1/h_B}$

③ Cylinder $Q = \left[\frac{T_i - T_o}{\frac{1}{h_i A_i} + \frac{\ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o A_o}} \right]$

④ $Q = U_o A_o \Delta T_{overall}$, $U_o = \frac{1}{\frac{A_o}{A_i} \frac{1}{h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o}}$

⑤ $Q = U_i A_i \Delta T_{overall}$, $U_i = \frac{1}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_o}{A_i} \frac{1}{h_o}}$

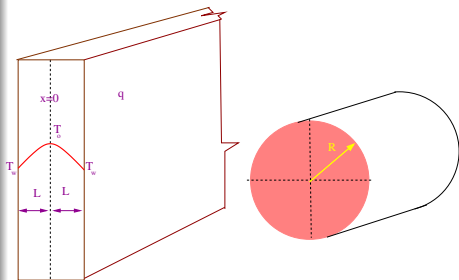
$$\frac{\ln(\omega/\omega_1)}{2k_1 H} \quad \frac{1}{2\omega_1 h_1 H}$$


$$\textcircled{2} \quad \frac{dq}{dr_o} = 0 = \frac{-2\pi L(T_i - T_\infty) \left(\frac{1}{kr_o} - \frac{1}{hr_o^2} \right)}{\left[\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h} \right]^2}$$

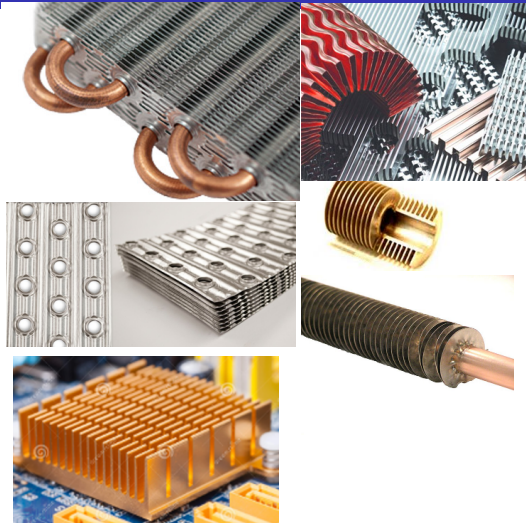
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Plane wall with heat source

- 1 $\frac{d^2 T}{dx^2} + \frac{q}{k} = 0$
- 2 Boundary conditions
 $T = T_w, \text{ at } x = \pm L$
- 3 Solution $\frac{T - T_o}{T_w - T_o} = \left(\frac{x}{L}\right)^2$
- 4 Alternatively $\frac{T - T_w}{T_o - T_w} = 1 - \left(\frac{x}{L}\right)^2$
- 5 Cylinder with heat generation
 $T = T_w, \text{ at } r = R$
- 6 Solution $\frac{T - T_w}{T_o - T_w} = 1 - \left(\frac{r}{R}\right)^2$



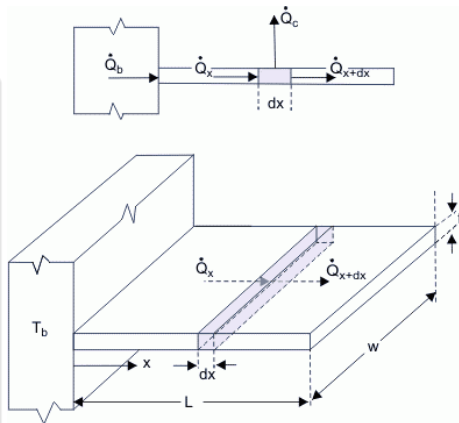
Extnded surfaces -Fins



- 1 Need for Extended surfaces
- 2 Applications

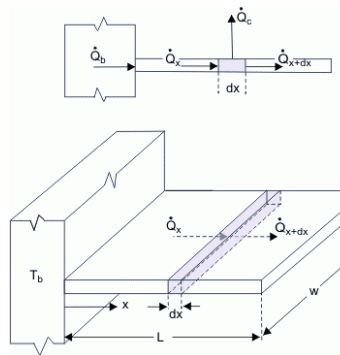
Heat transfer from fins

- 1 Base temperature T_b
- 2 Ambient temperature T_∞
- 3 Convection heat transfer coefficient $h(W/m^2K)$
- 4 Convective heat transfer $Q = hA_s(T - T_\infty)$
- 5 Surface perimeter P
- 6 Surface area $A_s = Pdx$
- 7 Energy balance



Heat transfer from fins

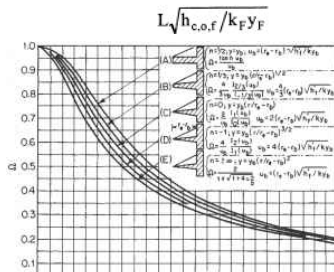
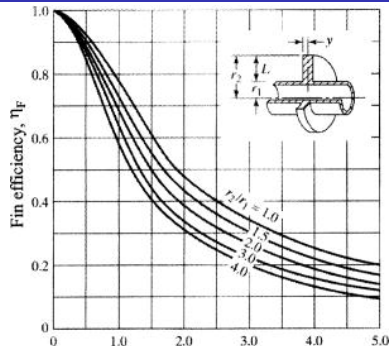
- 1 Ambient temperature T_∞
- 2 Surface perimeter P
- 3 Surface area $A_s = Pdx$
- 4 Heat transfer coefficient $h(W/m^2K)$
- 5 Convective heat transfer
 $Q = hA_s(T - T_\infty)$
- 6 Energy balance
- 7 Fin solution
 $\Theta = T - T_\infty = C_1e^{-mx} + C_2e^{mx}$
- 8 Fin very long
 $\Theta = \Theta_0 - at - x = 0, \Theta = 0 - at - x = \infty$
- 9 $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx}, m = \sqrt{\frac{hP}{kA}}$
- 10 Heat loss $q = -kA\frac{\partial T}{\partial x}|_{x=0} = \Theta_0\sqrt{hPkA}$



Fin Efficiency and effectiveness

- 1 Fin efficiency $-\eta_{fin} = \frac{Q - \text{Fin-actual}}{Q - \text{Fin-entire-fin-at-base-temp.}}$
- 2 Fin conduction resistance Vs convective resistance
- 3 Fin thickness varying - minimise weight
- 4 Fin effectiveness -

$$\varepsilon_{fin} = \frac{Q - \text{Fin-actual}}{Q - \text{with-out-fin-(base-area)}}$$
- 5 Fin effectiveness must be > 1



Steady state two dimensional

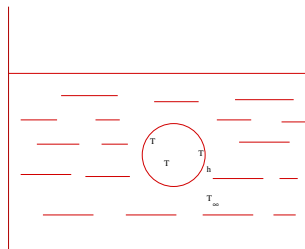
Two dimensional

- 1 $\frac{\partial^2(T)}{\partial x^2} + \frac{\partial^2(T)}{\partial y^2} = 0$
- 2 Heat flow $q_x = -kA_x \frac{\partial T}{\partial x}$
- 3 Heat flow $q_y = -kA_y \frac{\partial T}{\partial y}$
- 4 Two dimensional system with two temperatures $q = kS\Delta T_{overall}$
- 5 S- conduction shape factor

Lumped System Analysis

Assumptions

- 1 Interior temperature is uniform
- 2 One dimensional
- 3 h - constant
- 4 k - constant
- 5 h is low compared to k

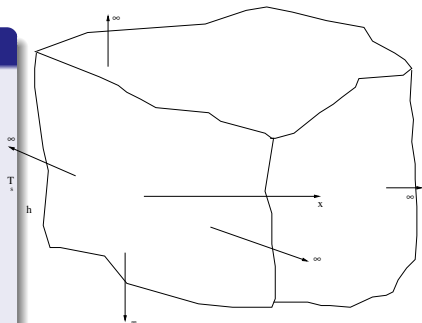


- ① Heat transfer by convection = Internal energy change
- ② $hA(T - T_{\infty}) = -mc_p dT/dt$
- ③ $\frac{d(T - T_{\infty})}{(T - T_{\infty})} = -\frac{hA_s dt}{\rho V c_p} = \frac{d\theta}{\theta} = -\frac{hA_s dt}{\rho V c_p}$
- ④ Limits $\theta = \theta_i \Rightarrow (T_i - T_{\infty}), t = 0$ and $\theta = \theta_t \Rightarrow (T - T_{\infty}), t = t$
- ⑤ integrating $\int_{\theta_i}^{\theta_t} \frac{d(\theta)}{\theta} = -\int_0^t \frac{hA_s dt}{\rho V c_p}$
- ⑥ $[ln(\theta)]_{\theta_i}^{\theta_t} = ln[\theta_t] - ln[\theta_i] = -\frac{hA_s t}{\rho V c_p}$
- ⑦ $ln\left[\frac{\theta_t}{\theta_i}\right] = -\frac{hA_s t}{\rho V c_p}$
- ⑧ $\frac{(T - T_{\infty})}{(T_i - T_{\infty})} = e^{-bt}$, where $b = \frac{hA_s}{\rho V c_p}$
- ⑨ Criterion Biot Number $Bi = hL/k \leq 0.1$, Error within 5%
- ⑩ $\frac{(T - T_{\infty})}{(T_i - T_{\infty})} = e^{BiFo}$, $Bi = \frac{hL}{k}$, $Fo = \frac{\alpha t}{L^2}$

Unsteady conduction semi infinite solids

Semi infinite solid

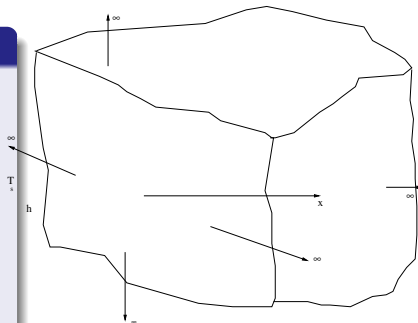
- 1 $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$
- 2 $T(0, t) = T_s, T(x \rightarrow \infty, t) = T_i$
- 3 initial condition $T(x, 0) = T_i$
- 4 Use similarity variable $\eta = \frac{x}{\sqrt{4\alpha t}}$
- 5 boundary condition $\eta = 0$, at $x = 0, \eta \rightarrow \infty$ at $x \rightarrow \infty$
- 6 after simplification $\frac{d^2 T}{d\eta^2} = -2\eta \frac{dT}{d\eta}$
- 7 $T(0) = T_s$ and $T(\eta \rightarrow \infty) = T_i$



Unsteady conduction semi infinite solids

Semi infinite solid contd

- 1 define a new variable $w = \frac{dT}{d\eta}$
- 2 $\frac{dw}{d\eta} = -2\eta w$
- 3 $\ln w = -\eta^2 + C_0$
- 4 $w = C_1 e^{-\eta^2}$, where, $C_1 = \ln C_0$
- 5 substituting $w = \frac{dT}{d\eta}$ and integrating
- 6 $T = C_1 \int_0^\eta e^{-u^2} du + C_2$
- 7 where 'u' is a dummy variable The boundary condition $\eta = 0$ gives $C_2 = T_s$



semi infinite solids- unsteady conduction

- ① applying boundary condition $\eta \rightarrow \infty$
- ② $T_i = C_1 \int_0^{\infty} e^{-u^2} du + C_2 = C_1 \frac{\sqrt{\pi}}{2} + T_s$
- ③ give $C_1 = \frac{2(T_i - T_s)}{\sqrt{\pi}}$
- ④ substituting C_1 and C_2
- ⑤ $\frac{T - T_s}{T_i - T_s} = \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du = \text{erf}(\eta) = 1 - \text{erfc}(\eta)$
- ⑥ where $\text{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du$ and $\text{erfc}(\eta) = 1 - \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du$
- ⑦ heat flux at the surface $q_s = -k \frac{\partial T}{\partial x} \big|_{x=0} = -k \frac{dT}{d\eta} \frac{\partial \eta}{\partial x} \big|_{x=0}$
- ⑧ $= -k C_1 e^{-\eta^2} \frac{1}{\sqrt{4\alpha t}} \big|_{\eta=0} = \frac{k(T_s - T_i)}{\sqrt{\pi \alpha t}}$

semi infinite solids- unsteady conduction

① specified surface temperature $T_s = C$

②
$$\frac{T(x,t)-T_i}{T_s-T_i} = \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

③
$$q(t) = \frac{k(T_s-T_i)}{\sqrt{\pi\alpha t}}$$

④ Specified surface heat flux $q_s = C$

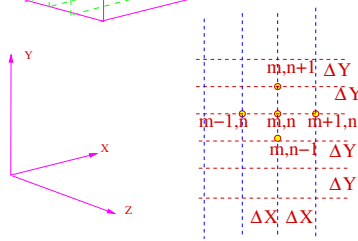
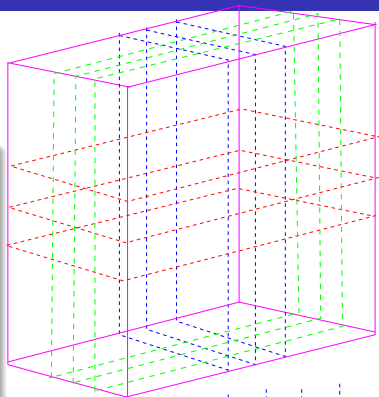
⑤
$$T(x,t) - T_i = \frac{q_s}{k} \left[\sqrt{\frac{4\alpha t}{\pi}} \exp\left(\frac{-x^2}{4\alpha t}\right) - x \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) \right]$$

⑥ convection on the surface $q_s(t) = h(T_\infty - T(0,t))$

⑦
$$\frac{T(x,t)-T_i}{T_\infty-T_i} = \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2\alpha t}{k^2}\right) \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right)$$

Numerical methods in conduction

- 1 Problem solutions $\Rightarrow$ Analytical, Experimental, Numerical
- 2 Numerical $\Rightarrow$ FDM, FEM, Finite Volume
- 3 FDM - Divide domain into nodes
- 4 Governing equations (PDE's) converted to algebraic equations
- 5 Write equations for each nodes



Numerical methods in conduction

① At the node $(m+1/2, n)$ $\frac{\partial T}{\partial x} = \frac{T_{m+1,n} - T_{m,n}}{\Delta x}$

② At the node $(m-1/2, n)$ $\frac{\partial T}{\partial x} = \frac{T_{m,n} - T_{m-1,n}}{\Delta x}$

③ At the node $(m, n+1/2)$ $\frac{\partial T}{\partial y} = \frac{T_{m,n+1} - T_{m,n}}{\Delta y}$

④ At the node $(m, n-1/2)$ $\frac{\partial T}{\partial y} = \frac{T_{m,n} - T_{m,n-1}}{\Delta y}$

⑤ At the node (m, n) $\frac{\partial^2 T}{\partial x^2} = \left[\frac{\left[\frac{\partial T}{\partial x} \right]_{m+1/2,n} - \left[\frac{\partial T}{\partial x} \right]_{m-1/2,n}}{\Delta x} \right]$

⑥ At the node (m, n) $\frac{\partial^2 T}{\partial x^2} = \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{\Delta x^2}$

⑦ At the node (m, n) $\frac{\partial^2 T}{\partial y^2} = \left[\frac{\left[\frac{\partial T}{\partial y} \right]_{m,n+1/2} - \left[\frac{\partial T}{\partial y} \right]_{m,n-1/2}}{\Delta y} \right]$

⑧ At the node (m, n) $\frac{\partial^2 T}{\partial y^2} = \frac{T_{m,n+1} + T_{m,n-1} - 2T_{m,n}}{\Delta y^2}$

⑨ 2D equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{\Delta x^2} + \frac{T_{m,n+1} + T_{m,n-1} - 2T_{m,n}}{\Delta y^2} = 0$$

⑩ Solution - Gauss - Seidel Iteration etc.